$= x_3 \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix} + \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$ The solution set is the line through the vector $\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$ parallel to the vector $\begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$.

(Scalar multiplication)

The vector $x_3 \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$ is the solution set of Exercise 5.

17. Describe and compare the solution sets of $x_1 + 9x_2 - 4x_3 = 0$ and $x_1 + 9x_2 - 4x_3 = -2$.

$\boxed{\text{Solution:}}$ $x_1[1] + x_2[9] + x_3[-4] = [-2]$.

By Defn. of $Ax = b$, we have: $\begin{bmatrix} 1 & 9 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = [-2]$

By Theorem 3, the matrix equation $Ax = b$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \; b]$. $[1 \; 9 \; -4 \; -2]$. Verification of whether the matrix is in rref

1. Row 1, being a non-zero row and the only row makes the condition that all non-zero rows must be above the zero rows vacuously True.
2. Row 1 Leading element $= 1$ in Column 1

$\therefore$ The condition that for each row, the leading element must be in a column to the right of the columns of the leading elements of all rows above it is ~~True~~ vacuously True.

3. The condition that for each column having a leading element, all positions below it must contain 0 is vacuously True.
4. The condition that each leading element must be 1 is True.
5. The condition that for each column having a leading element, all positions above it must contain 0 is vacuously True.

$\therefore$ The matrix $[1 \; 9 \; -4 \; -2]$ is in rref $\square$.

By defn, Columns 2 and 3 are not pivot columns. $\therefore$ By defn, x_2 and x_3 are free variables. $\therefore x_1 + 9x_2 - 4x_3 = -2 \Rightarrow x_1 = -9x_2 + 4x_3 - 2$